

## Topic 19: Resolution, Sensitivity & accuracy specification

### → Resolution in digital multimeters

The no. of digit positions (or simply the no. of digits) used in digital meters determines the resolution. Thus a 3 digit display on a DVM for 0-1 V range will be able to indicate values from 0-999 mV with smallest increment of "resolution" of 1 mV.

The resolution of a digital meter is determined by the no. of active or full digits used.

i)  $n = \text{no. of full digits}$

$$\Rightarrow \boxed{\text{Resolution} = 1/10^n}$$

### → Sensitivity of digital meters

sensitivity is the smallest change in input which a digital meter is able to detect.

Hence, Sensitivity,  $S = (f.s.)_{\min} \times R$

Here  $(f.s.)_{\min} = \text{lowest full scale value of meter}$

$R = \text{resolution expressed as decimal.}$

### → Accuracy specifications for digital meters

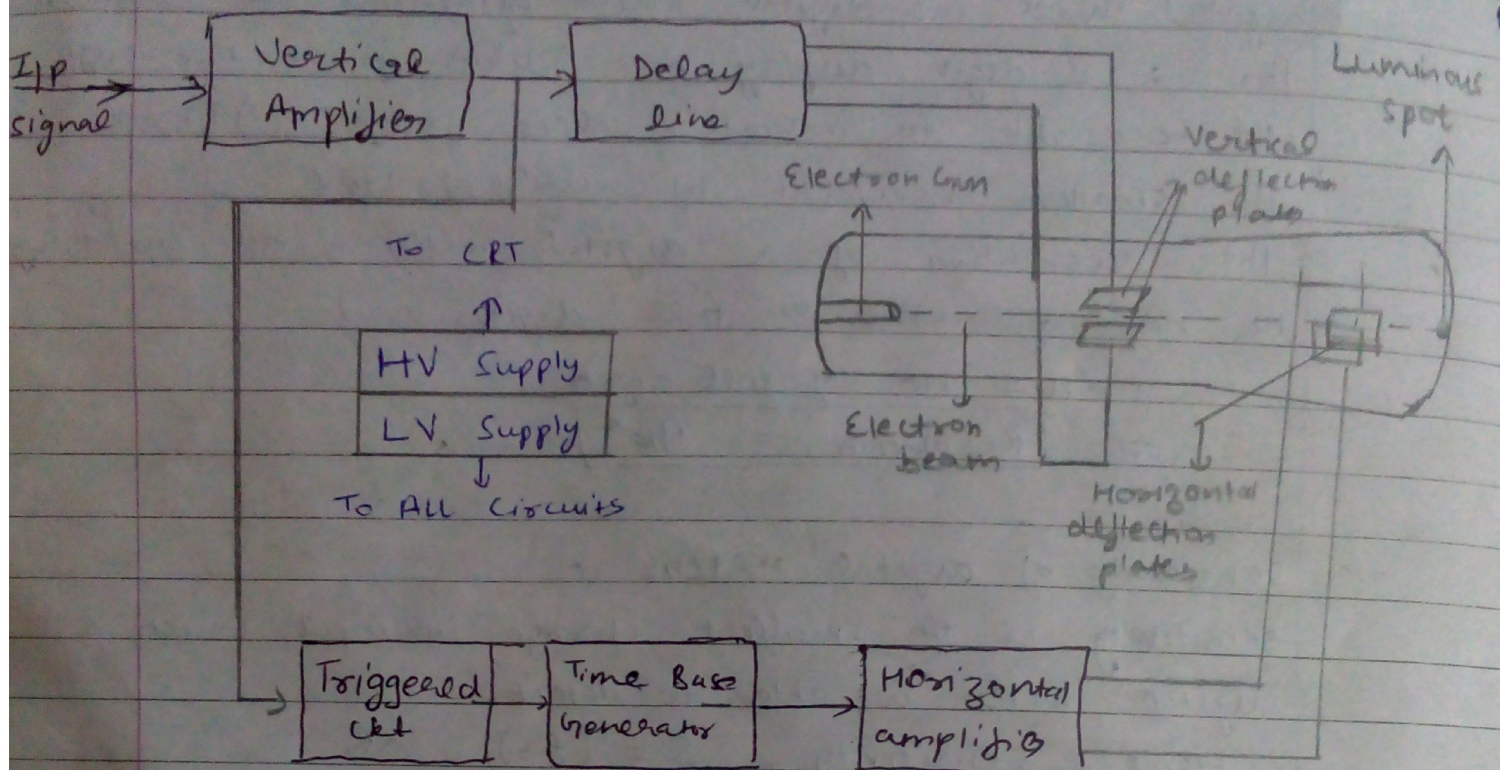
Accuracy is usually expressed as % of reading plus a % of full scale. Often the x. age of full scale part of the statement is given in terms of digits.

When this is done, the digits referred to are the least significant digit (LSD)



→ CRO (Cathode Ray oscilloscope)

Cathode ray oscilloscope is an instrument which we use to display, measure and analyse various signals. CRO is very fast X-Y plotter that can display input signal v/s time or other signals.



BLOCK DIAGRAM OF CRO

### \* WORKING OF A CRO

→ A CRO uses luminous spot which is used by striking beam of electrons and this luminous spot moves in response variation in the input quantity.

→ The instrument employs a cathode ray tube (CRT) which generates the electron beam, accelerates the beam to high velocity, deflects the beam to create image & contains a phosphor screen where the  $e^-$  beam eventually becomes visible.

→ Low voltage supply is required for heating of  $e^-$  gun bar.



generation of electron beam. High voltage is required for CRT to accelerate the beam.

- Horizontal and vertical deflection plates deflect the beam acc. to i/p signal.
- $e^-$  beam strikes the screen & creates a visible spot. This spot is deflected on the screen in horizontal dir<sup>n</sup> with constant time dependent rate which is accomplished by a Time Base Circuit in oscilloscope.
- The signal to be viewed is supplied to vertical deflection plates through vertical amplifier which raises the potential of i/p signal to a level that will provide usable deflection of  $e^-$  beam.

TRIGGERING  
of  
CRO

- Now  $e^-$  beam deflects in Z directions, horizontal on X axis & vertical on Y axis. A TRIGGERING CIRCUIT is provided for synchronizing two types of deflection so that horizontal deflection starts at the same point of input vertical signal each time it sweeps.

### Applications of CRO

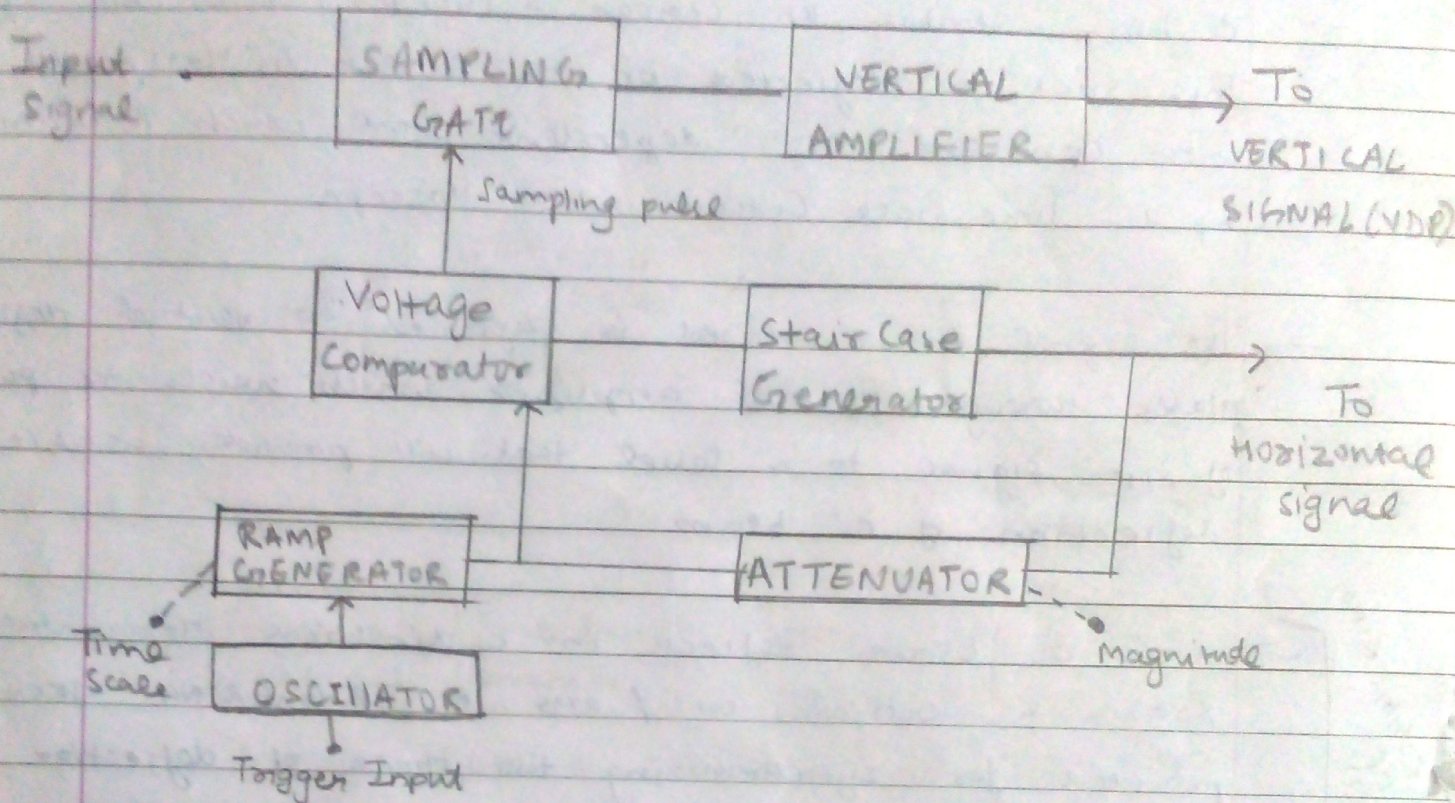
- Measurement of voltage & current - Voltage waveform will be made on oscilloscope screen, voltage can be measured by seeing its amplitude variation on screen. Similarly current waveform can be read by measuring peak to peak maximum current value on the screen.
- measurement of phase - Phase measurement in CRO can be done by help of Lissajous pattern figures which tell us about the phase difference between two signals.



→ Measurement of frequency → frequency can be measured using a CRO as well with the help of measuring the time period of the signal whose frequency is to be calculated.

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## SAMPLING OSCILLOSCOPE



→ A sampling oscilloscope is used to examine very fast signals. It is similar in principle to the use of stroboscopic light to look at fast mechanical motion.

Samples are taken at different portions of the waveform over successive cycles and then the total picture is stretched, amplified by relatively low bandwidth amplifiers and displayed by a continuous wave on screen.

→ The advantage of sampling oscilloscope is that it can measure very high speed sweeps which require sweep speeds of order  $10^9$  ps per division and amplifiers bandwidths



of 15 GHz. using instruments of lower bandwidths  
 \* Role of Delay in Sampling Oscilloscope

- A delay sweep feature is used in many laboratory purpose oscilloscope. By using this feature in instrument it increases the versatility by selection of position of undelayed sweep, measure waveform jitter or noise time and check pulse time modulation.
- Hence delayed sweep is a technique that adds precise amount of time between the trigger point and beginning of scope sweep.

## Topic 21% RECORDERS

- \* **Recorders** - It is often necessary to have a permanent record of the state of phenomenon being investigated. A recorder thus records electrical and non-electrical quantities as a function of time. Current and voltage can be recorded directly whereas non-electrical quantities are first converted to equivalent currents and voltages with the help of sensors and transducers.
- \* **Recorder Requirements** - One of the important considerations in an instrumentation system is the method by which the data acquired is recorded. The recording method should be consistent with the type of system. If we are dealing with a wholly analog system then analog recording techniques should be used while on the other hand if the system has



digital output, digital recording devices are used. Thus there are two types of recording devices -

- (i) Analog Recorders
- (ii) Digital Recorders.

Analog Recorders → They can be broadly classified into -

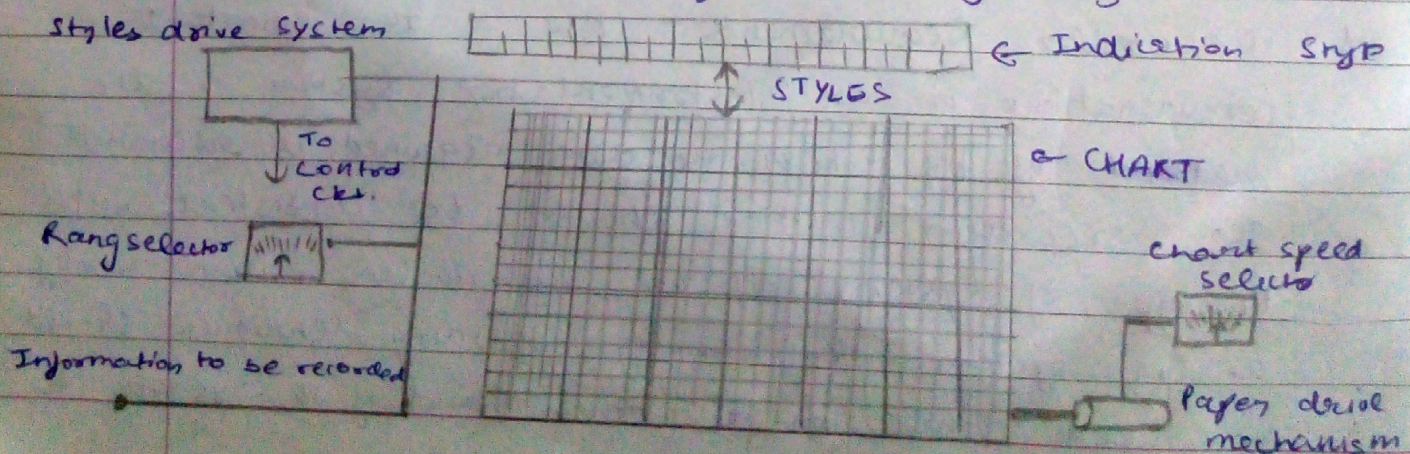
- (i) Graphic Recorders
- (ii) Oscillographic recorders
- (iii) Magnetic tape recorders

Graphic recorders → They are devices which display and store a pen and ink record of history of some physical event. They are divided into 2 categories -

1. Strip chart recorders - Records one or more variables w.r.t time. It is an X-t recorder.
2. X-Y Recorders - Records one or more dependent variables w.r.t an independent variable.

Strip Chart Recorder - It consists of -

1. A long roll of graph paper moving vertically
2. A system for driving the paper at some selected speed
3. A stylus for making marks on moving graph paper. It moves horizontally in proportion to quantity being recorded
4. A stylus driving system which moves the stylus in a nearly exact replica or analog of quantity being recorded

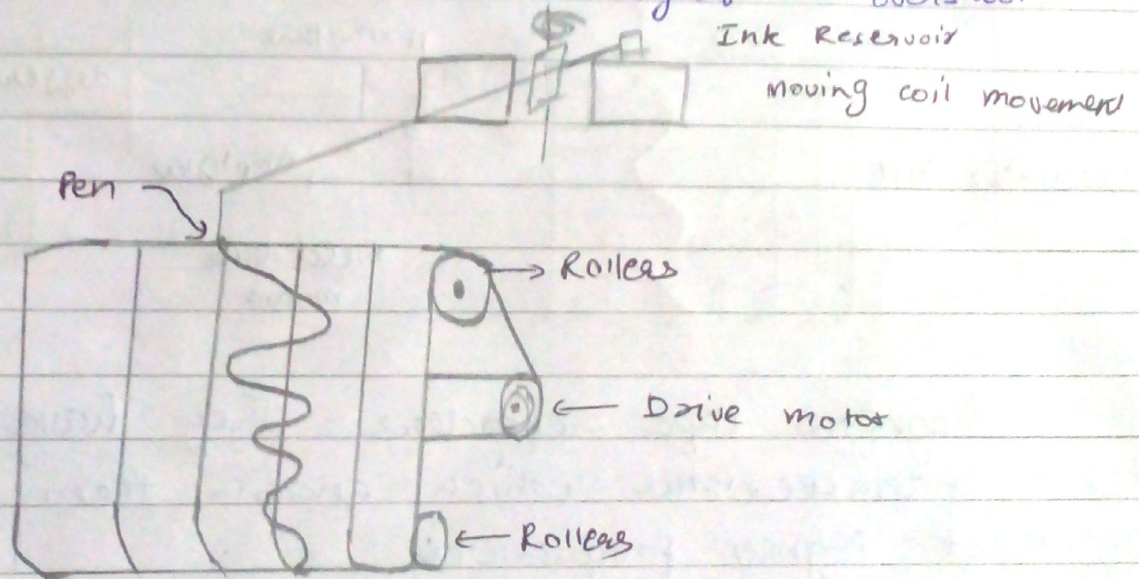




→ Types of Strip chart recorders -

→ GALVANOMETER TYPE - This type of strip chart recorder operates on deflection principle. The deflection is produced by a galvanometer which produces a torque on account of a current passing through coil. This current is proportional to the quantity being measured.

→ This type of recorder only records average values. It can not record fast variations in either current, voltage or power. The instrument is to be critically damped so that there is no significant overshoot.



Components of galvanometer type Recorder

→ Null Type Recorders - This type of recorder operates on comparison basis.

A change in the i/p produced by signal from sensor or transducer upsets the balance of measuring circuit of recorder. As a result of this unbalance an error signal is produced that operates some device which restores balance or bring it to <sup>Null</sup> ~~Null~~ conditions. This amount of movement is indication of magnitude and direction of



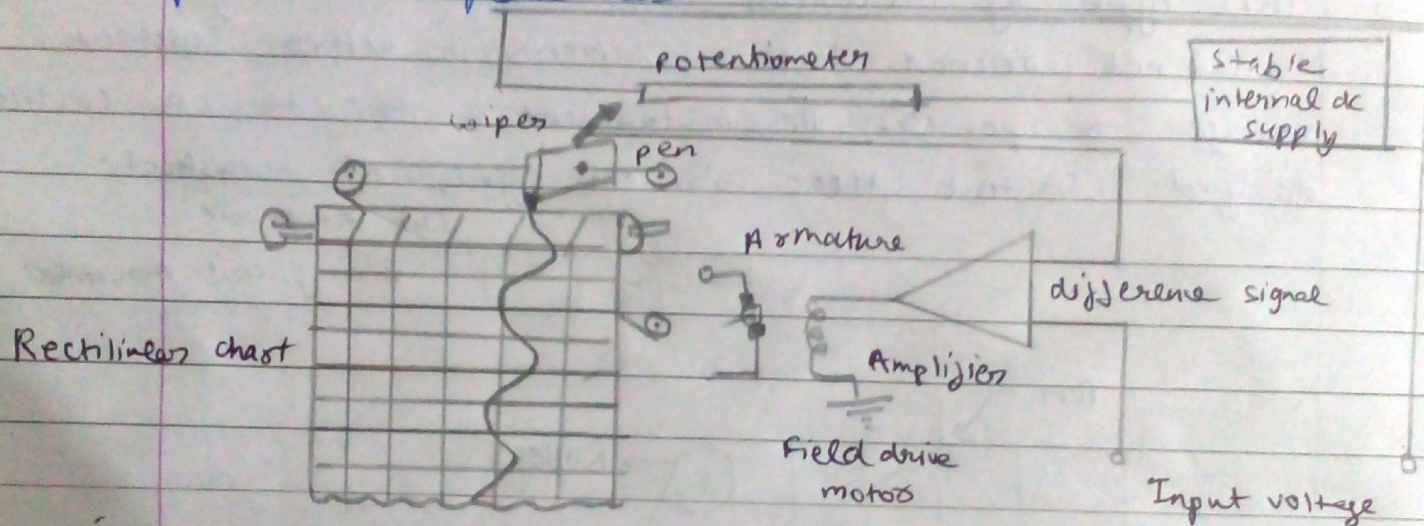
the quantity being measured has deviated from Normal  
Various types of null recorders are

- (i) Potential Recorder (ii) Bridge recorders (iii) LVDT recorders

The principle

**Potential Recorder** — The principle of operation of all the above null recorders are same i.e. to obtain null conditions.

→ The ~~disadvantage~~ most common application of potentiometer recorder is for recording and control of process temperatures.



**Magnetic tape recorders** — They have response characteristics which enable them to be used at higher frequencies.

They have low distortion. The magnitude of electrical i/p signal is stored in magnetic memory and this signal can be produced whenever required.

- When the information has been processed, the tape can be erased and reused to record a new set of data.

